

BUILDING ROBUST LANGUAGE MODEL SYSTEMS: INTROSPECTION+UPDATABILITY+UNCERTAINTY



Reexpress AI
<https://re.express/>

Language model (LM) agents are **unreliable, uninterpretable, and overconfident**.

Solution: Model neural networks as decomposable, hidden instance-based metric learners via **Similarity-Distance-Magnitude (SDM) activations**.

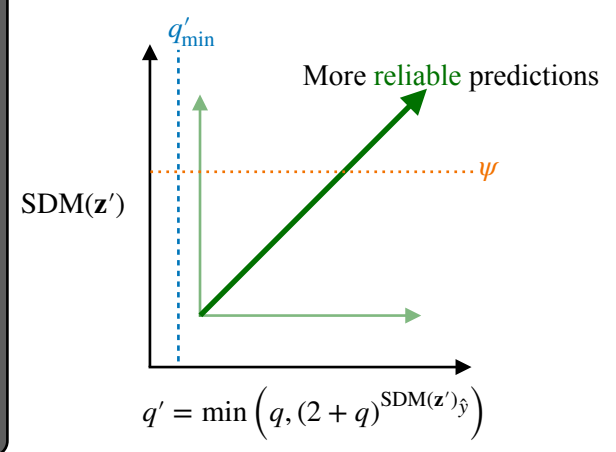
Enables **ACTIONABLE INTERPRETABILITY**:

INTROSPECTION of the predictions against instances with known labels (*interpretability-by-exemplar*)

Local **UPDATABILITY** of conditional-branching decisions without a full re-training (as a mechanism complementary to in-context learning, retrieval, and tool-calling)

Robust estimators of the predictive **UNCERTAINTY**

SDM estimator. Data-driven partitioning of the calibration set (via an SDM activation) to estimate a region with class- and prediction-conditional accuracy at least α , a value near 1.

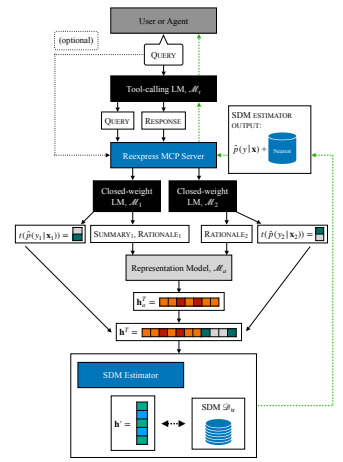


Criteria for selective classification:

$$\text{SDM}_{\text{HR}} := \begin{cases} \hat{y} & \text{if } q' \geq q'_{\min} \wedge \text{sdm}(z')_{\hat{y}} \geq \psi_{\hat{y}} \\ \perp & \text{otherwise} \end{cases}$$

SDM estimators are the substrate for building robust language model systems

The **REEXPRESS MCP SERVER** provides $\hat{p}(y | \mathbf{x})$, a robust, interpretable estimate of instruction-following of a tool-calling LM, $\mathcal{M}_t$, via an SDM estimator.



Binary classification of instruction-following: Verbalized uncertainty is unreliable, while Reexpress MCP Server's estimates are introspectable, locally updatable, and well-calibrated

$\alpha = 0.9$; Class 1: NOT Verified; Class 2: Verified		y = 1		y = 2		y ∈ {1,2}	
Dataset	Method	Acc.	$\frac{n}{ \mathcal{D}_{\text{te}} }$	Acc.	$\frac{n}{ \mathcal{D}_{\text{te}} }$	Acc.	$\frac{n}{ \mathcal{D}_{\text{te}} }$
HLE-TEST	GPT5.5+GEMINI3.1 VERBALIZED PROBABILITY	0.05	0.19	0.98	0.39	0.68	0.57
OPENTHOUGHTS-TEST	GPT5.5+GEMINI3.1 VERBALIZED PROBABILITY	0.95	0.48	0.88	0.44	0.92	0.92
MATHNET-TEST	GPT5.5+GEMINI3.1 VERBALIZED PROBABILITY	0.77	0.39	0.92	0.45	0.85	0.84
HLE-TEST	REEXPRESSV2.4.0 SDM ESTIMATOR (LOWER)	0*	<0.01	1	0.01	0.83*	0.01
OPENTHOUGHTS-TEST	REEXPRESSV2.4.0 SDM ESTIMATOR (LOWER)	0.96	0.34	0.98	0.30	0.97	0.64
MATHNET-TEST	REEXPRESSV2.4.0 SDM ESTIMATOR (LOWER)	0.96	0.21	0.94	0.17	0.95	0.38

*The one incorrect prediction corresponds to a ground-truth label annotation error.

Dense vector matching for LMs.

RAG uses dense matching for information retrieval (in *semantic-representation* space).

SDM activations use dense matching for uncertainty quantification and interpretability (in *feature-representation* space).



Foundation Paper



System Paper



MCP Server

SDM Activation: A Memory Layer for Actionable Interpretability

Simply add a 1-D CNN+linear layer and replace the final-layer SOFTMAX

$$\text{SDM}(z')_i = \frac{\text{Similarity}^{\text{Distance} \cdot \text{Magnitude}_i}}{\sum_{c=1}^C \text{Similarity}^{\text{Distance} \cdot \text{Magnitude}_c}} = \frac{(2 + q)^{d \cdot z'_i}}{\sum_{c=1}^C (2 + q)^{d \cdot z'_c}}, 1 \leq i \leq C$$

Distilled representations from the 1-D CNN adaptor are used for dense matching:

d : L^2 **Distance** to nearest training/support instance 1-quantile (lower $L^2 \rightarrow$ higher d).

q : **Similarity**. Count of consecutive (depth-wise) nearest matches in training/support that are correctly predicted *and* match $\hat{y}$ of the test instance.